

6	(a)	It is the force experienced per unit length of wire carrying per unit electric current when placed inside a magnetic field, with the conductor placed perpendicular to the magnetic field.	
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	(b)	(i)	$B_{WX, P} = \frac{\mu_0 I_{WX}}{2\pi r_p}$ $= \frac{(4\pi \times 10^{-7})(9.0)}{2\pi(50 \times 10^{-3})}$ $= 3.6 \times 10^{-5} \text{ T}$	
		(ii)	$B_{YZ, P} = \frac{\mu_0 I_{YZ}}{2\pi r_p}$ $= \frac{(4\pi \times 10^{-7})(12.0)}{2\pi(50 \times 10^{-3})} = 4.8 \times 10^{-5} \text{ T}$ $B_{res} = \sqrt{B_{WX, P}^2 + B_{YZ, P}^2}$ $= \sqrt{(3.6 \times 10^{-5})^2 + (4.8 \times 10^{-5})^2}$ $= 6.0 \times 10^{-5} \text{ T}$	

7	(c)	(i)	As the frequency increases, the inductive reactance increases, and the capacitive reactance decreases.	
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